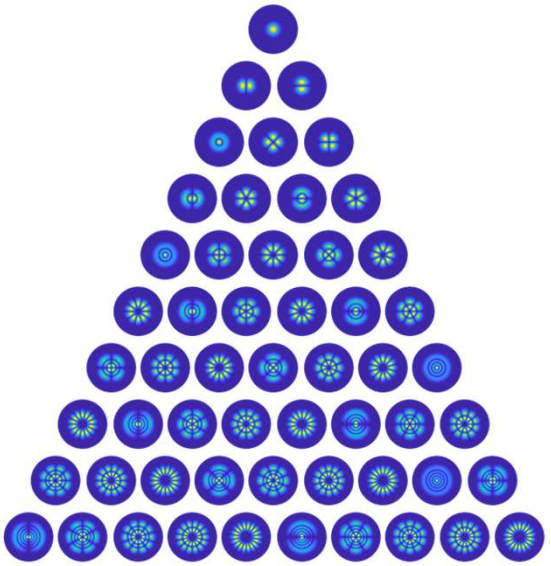
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# Advanced optical fiber communications

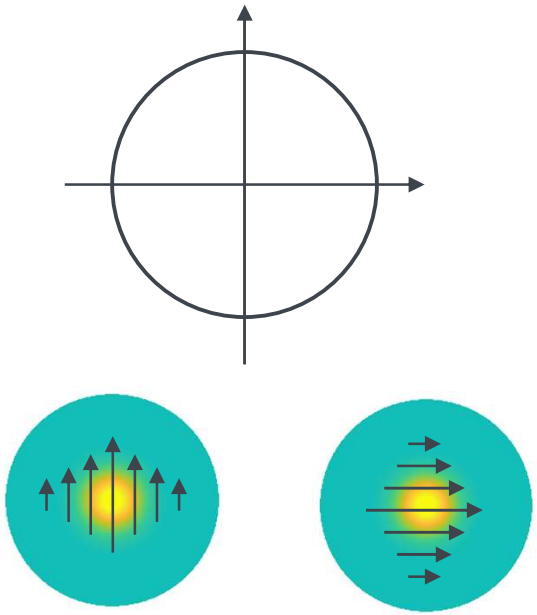
## Wintersemester 2025/26

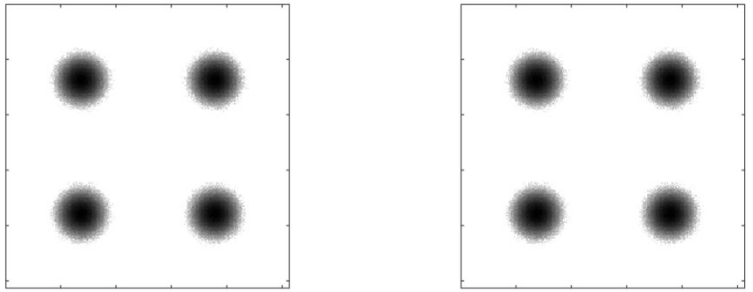
## Lecture 13



Prof. Dr.-Ing. Georg Rademacher

# Polarization multiplex (Pol Mux)





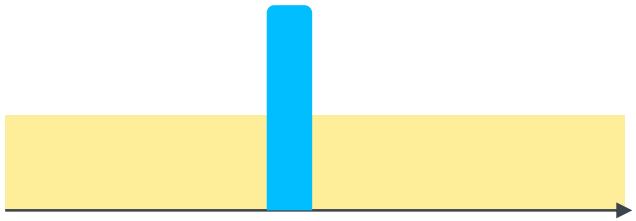
# OSNR in long distance transmission



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# SNR vs. OSNR

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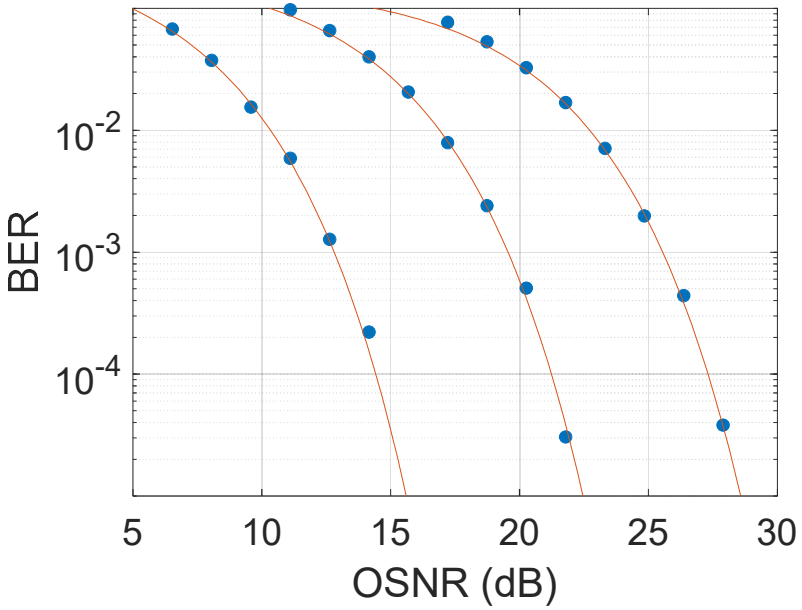


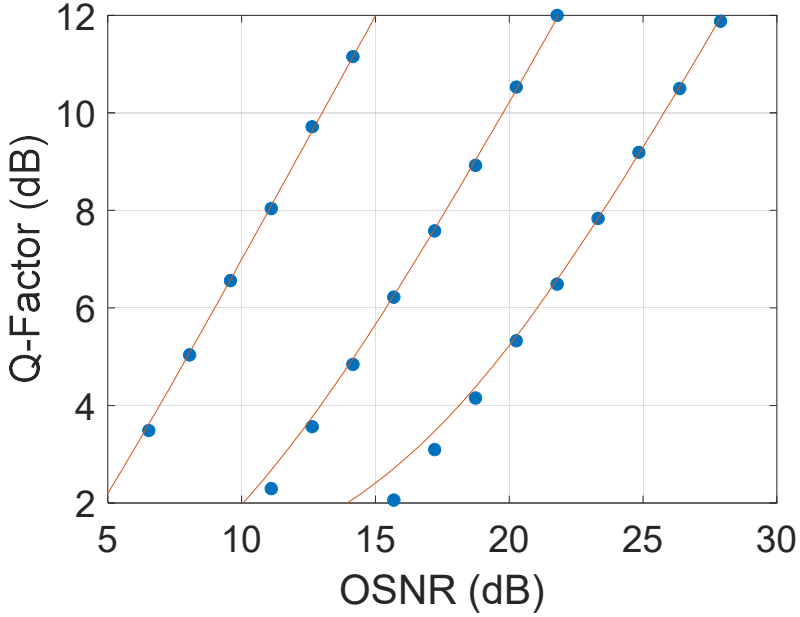


# Coherent Signals

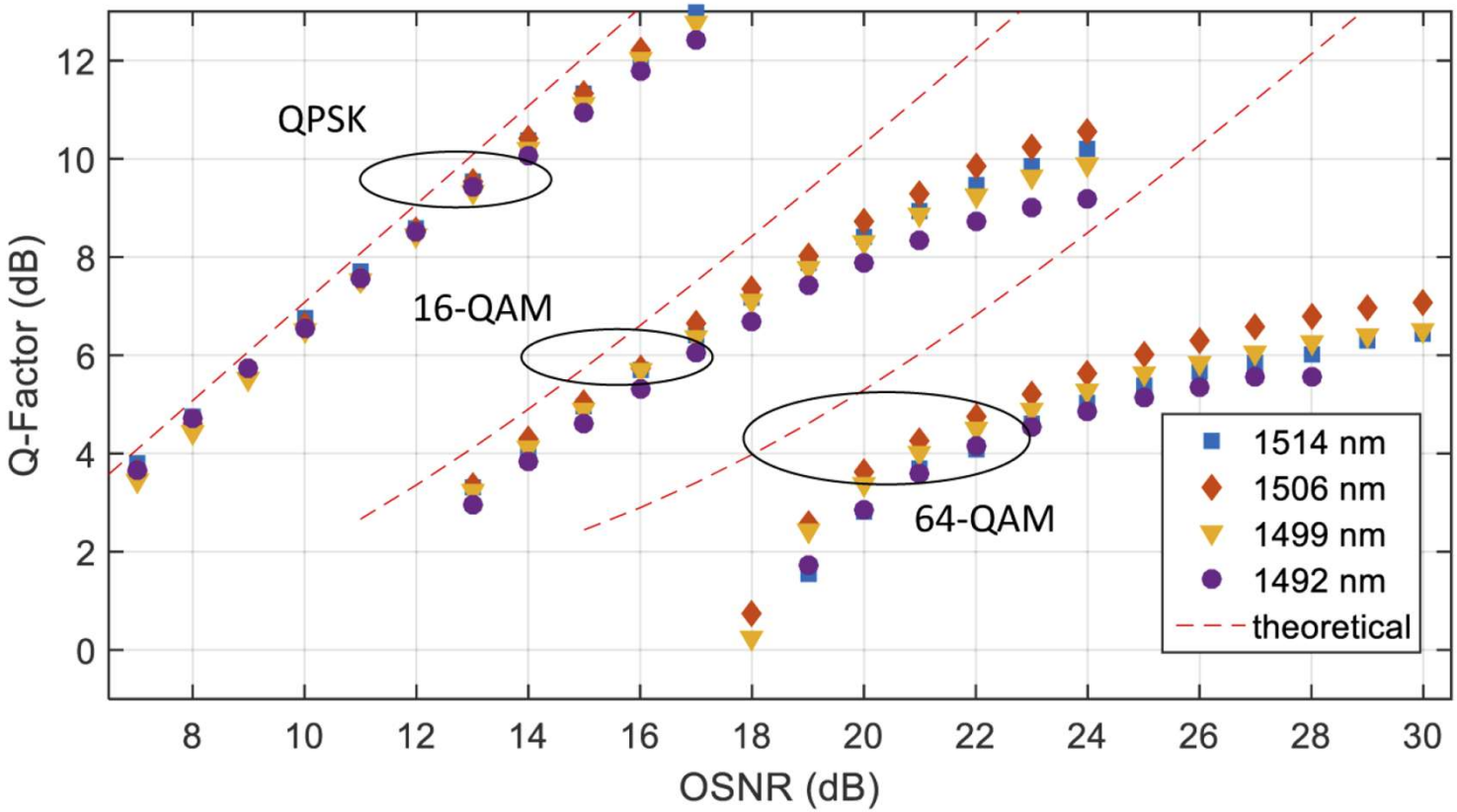
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Inverse complementary error function



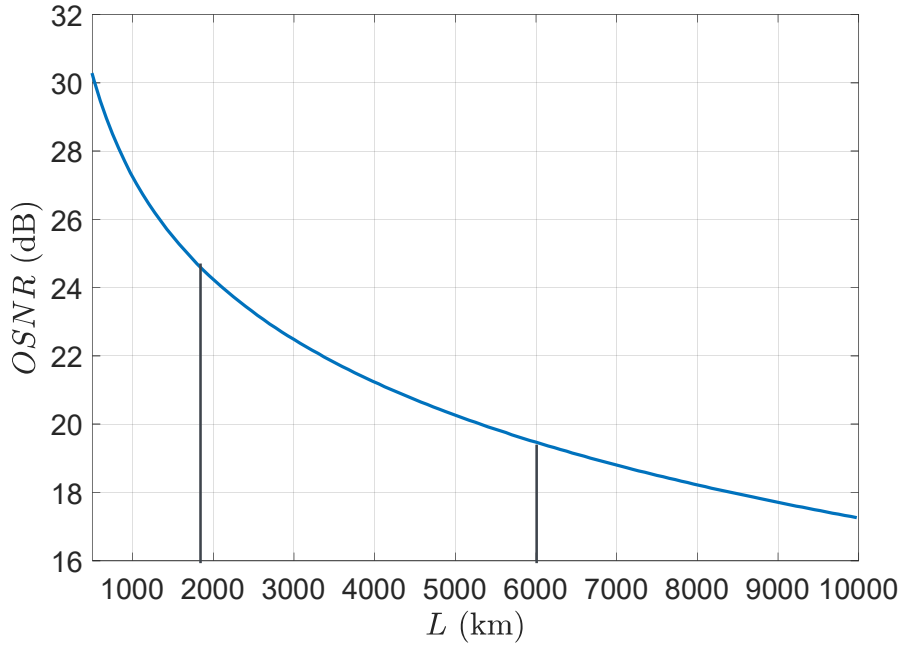



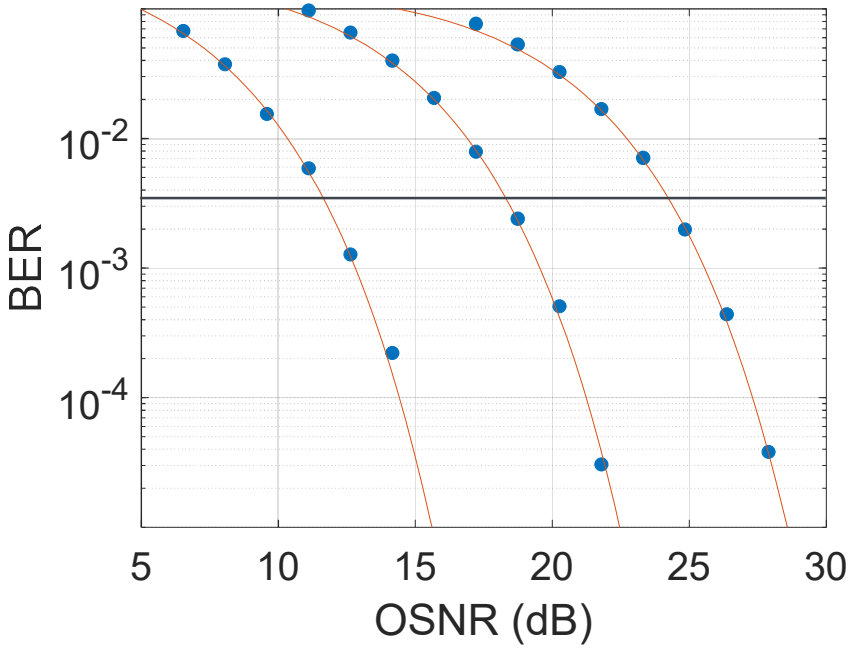
# Measured performance



# Long distance transmission

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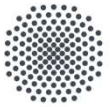



# Pulse propagation in SMF



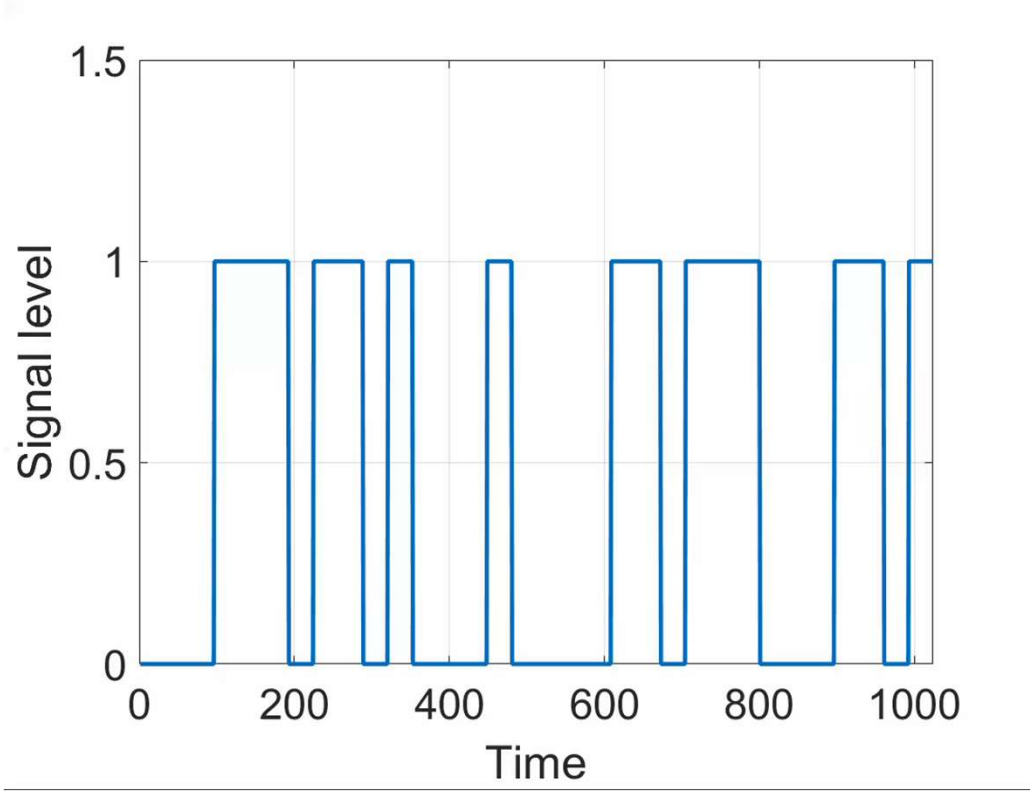
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# Pulse propagation in SMF



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# Pulse propagation in SMF: Chromatic dispersion

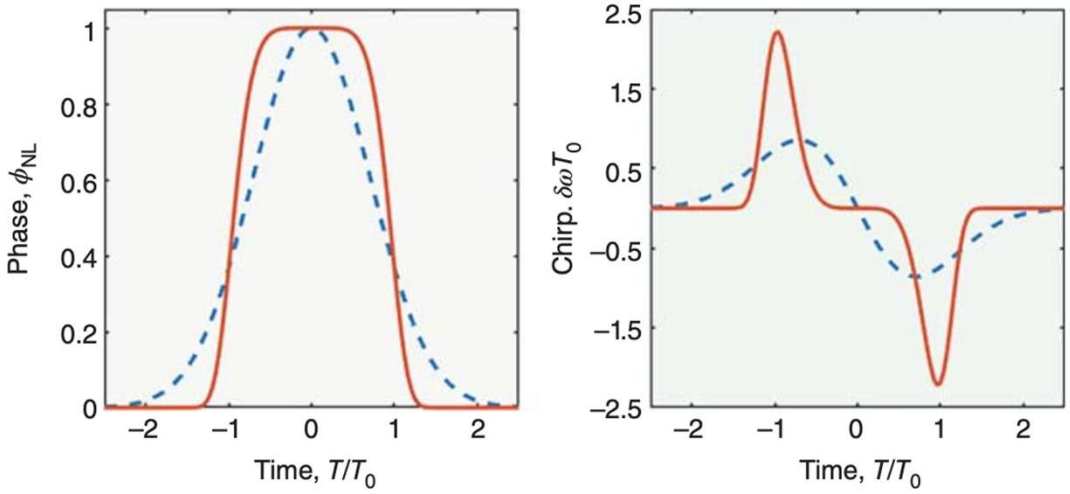


# Nonlinear signal distortions

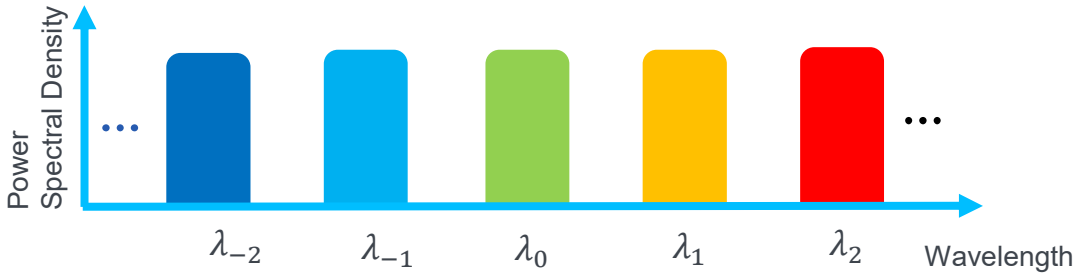


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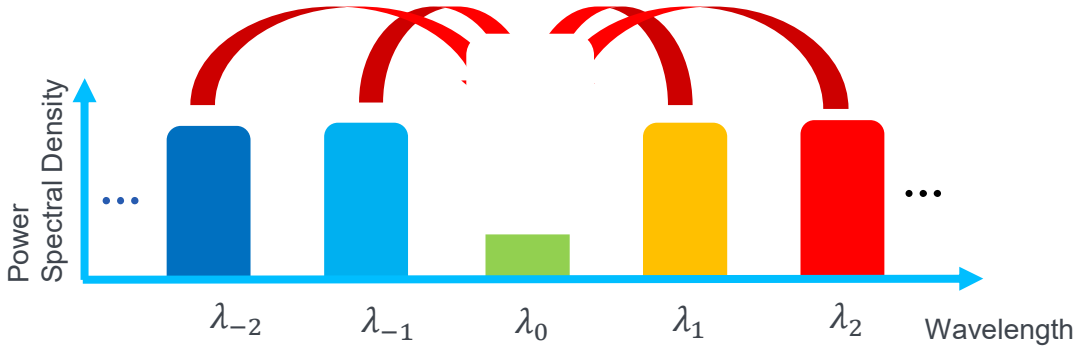
# Self-phase modulation

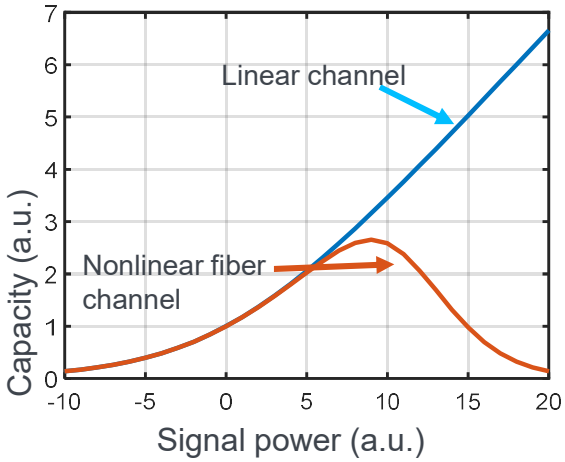


# Cross-phase modulation



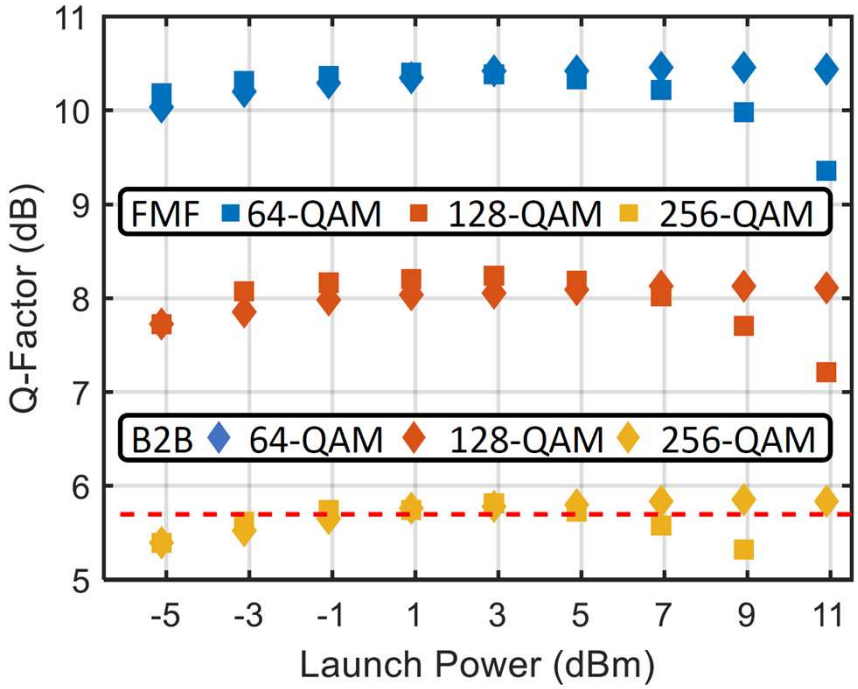
# Nonlinear signal distortions



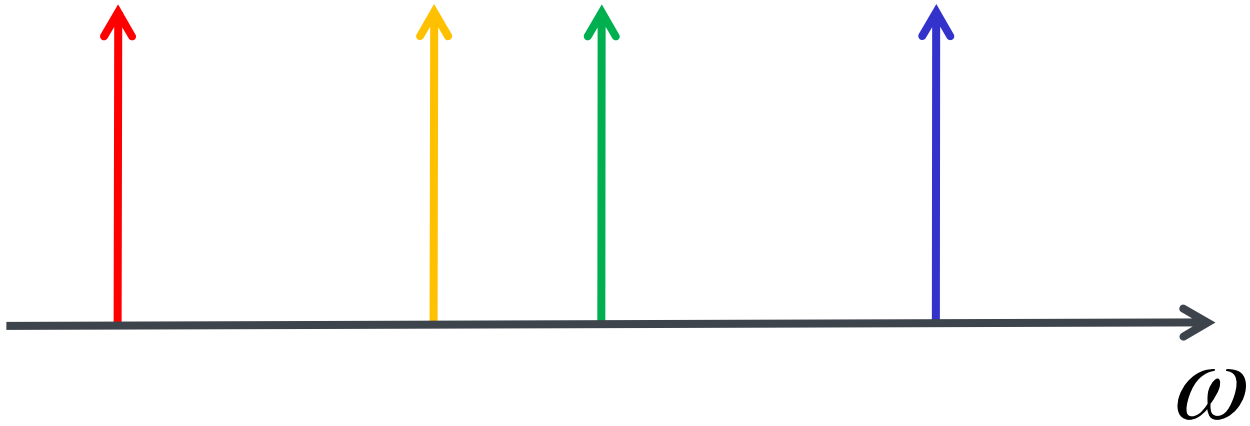
# Nonlinear signal distortions

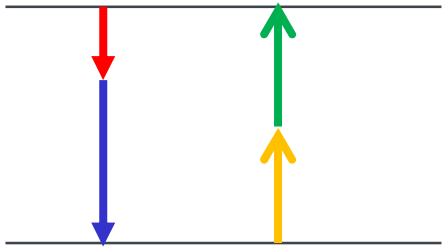
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# Four-wave mixing

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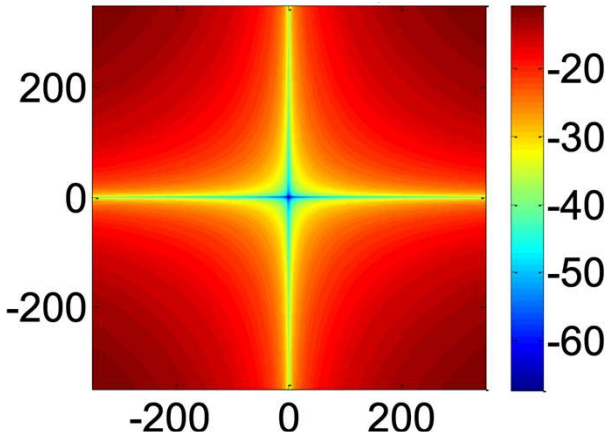


# Four-wave mixing

# Four-wave mixing

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# Nonlinear optical signal-to-noise ratio



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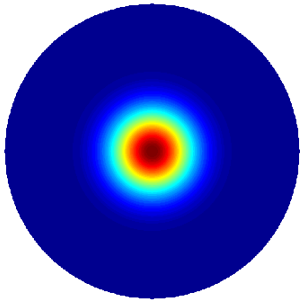
# The GN Model of Non-Linear Propagation in Uncompensated Coherent Optical Systems

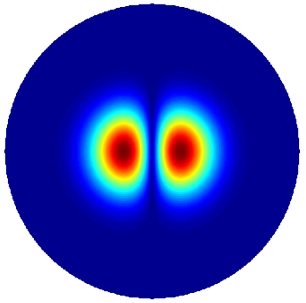
Pierluigi Poggiolini, *Member, IEEE*

*(Invited Paper)*

# Four-wave mixing

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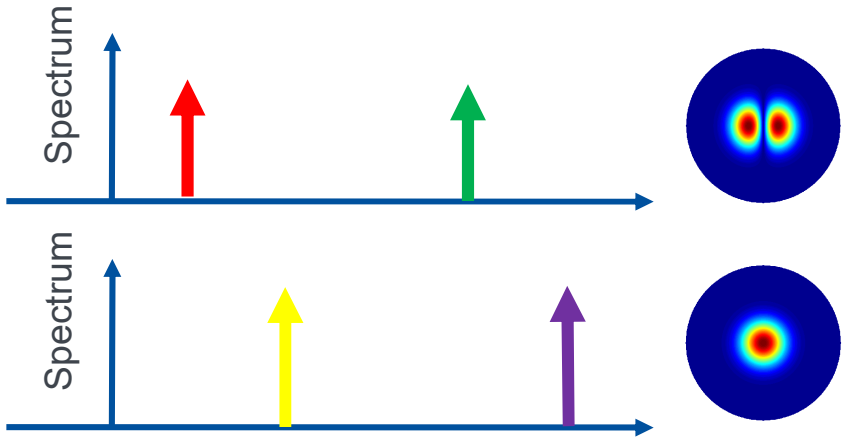


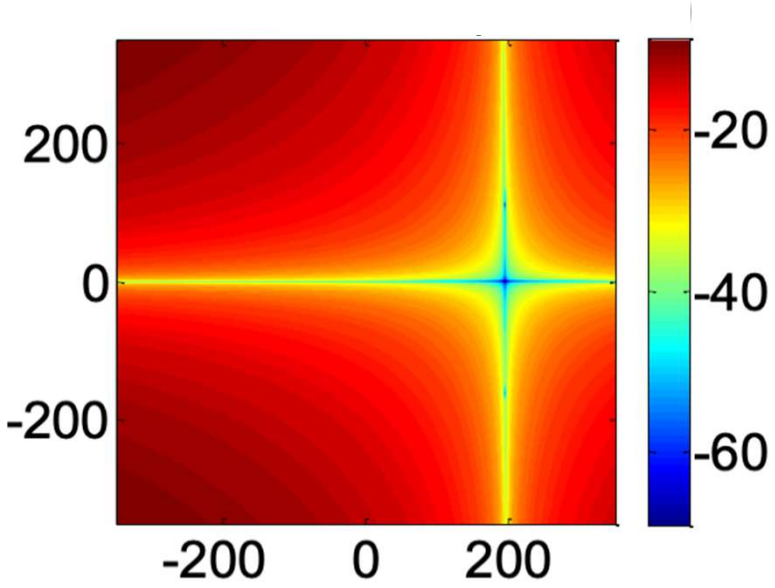


# Intermodal four-wave mixing

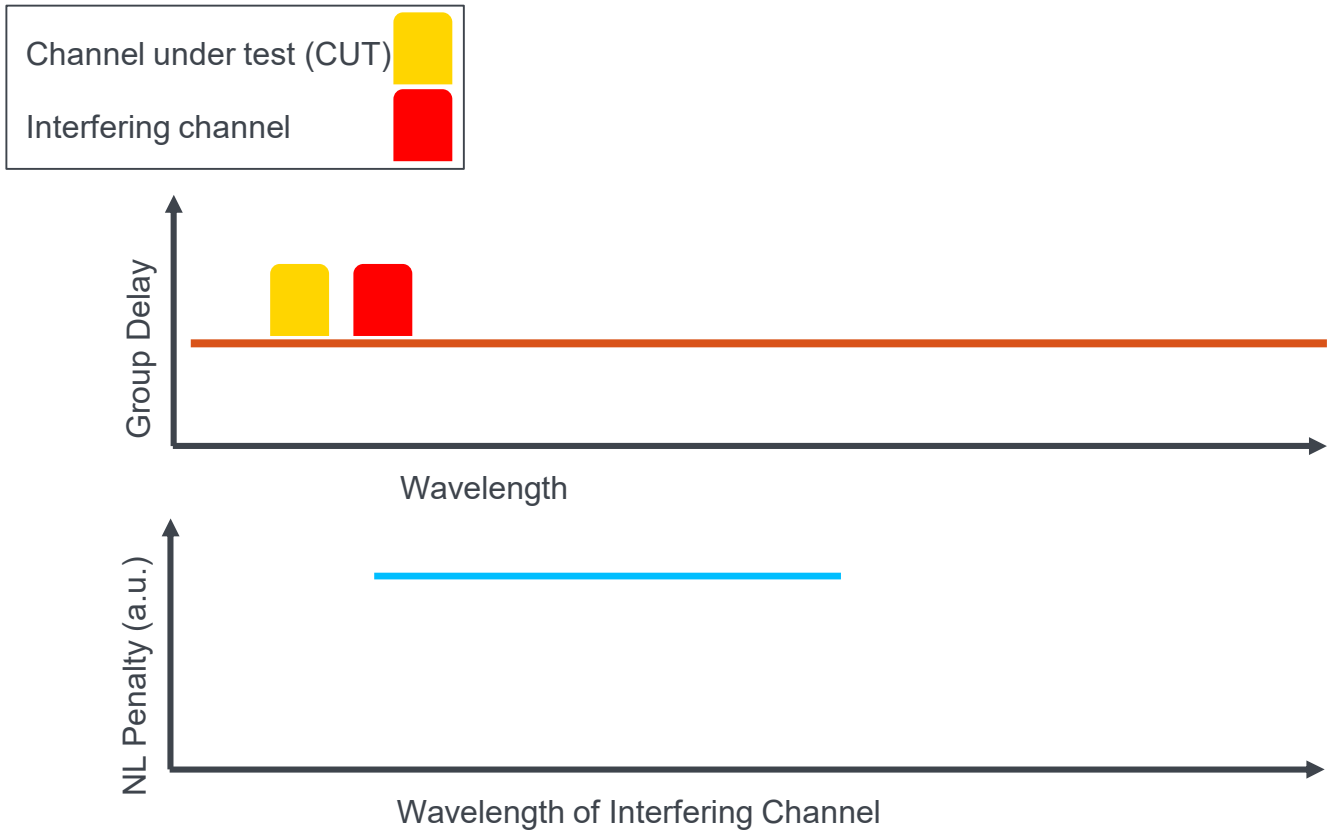
# Intermodal four-wave mixing



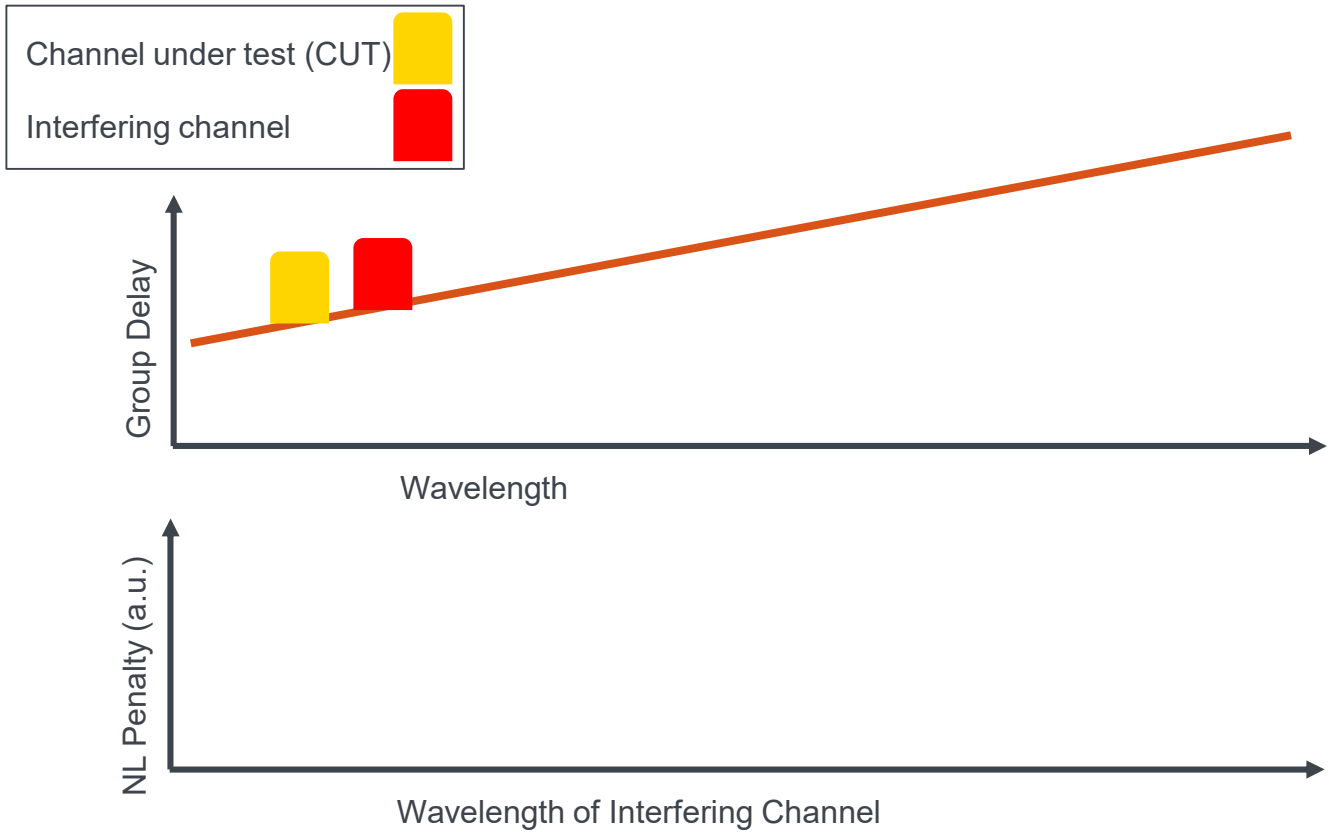


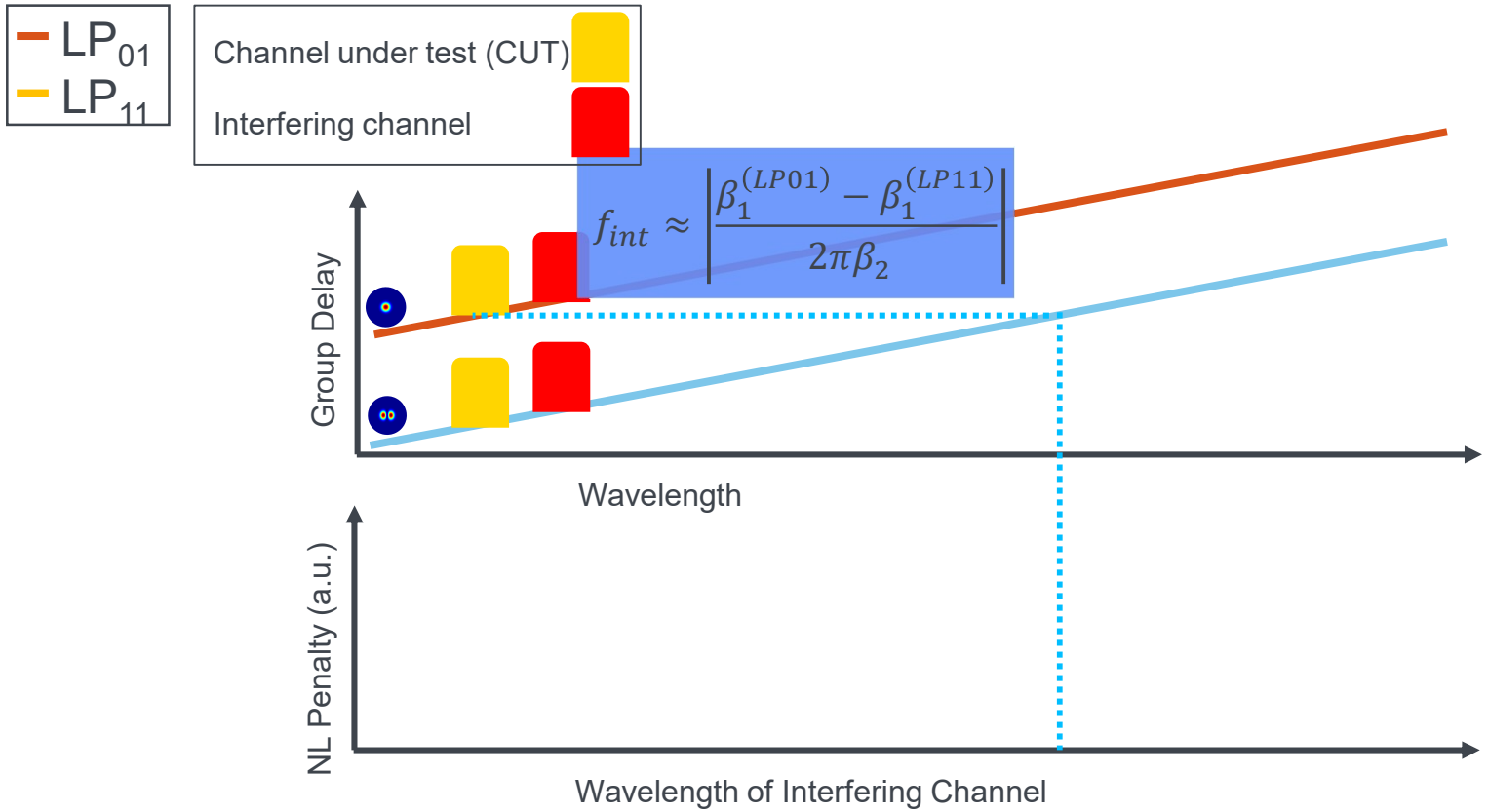
# Nichtlineare Signalverzerrung in einer nicht dispersiven Single-Mode Faser



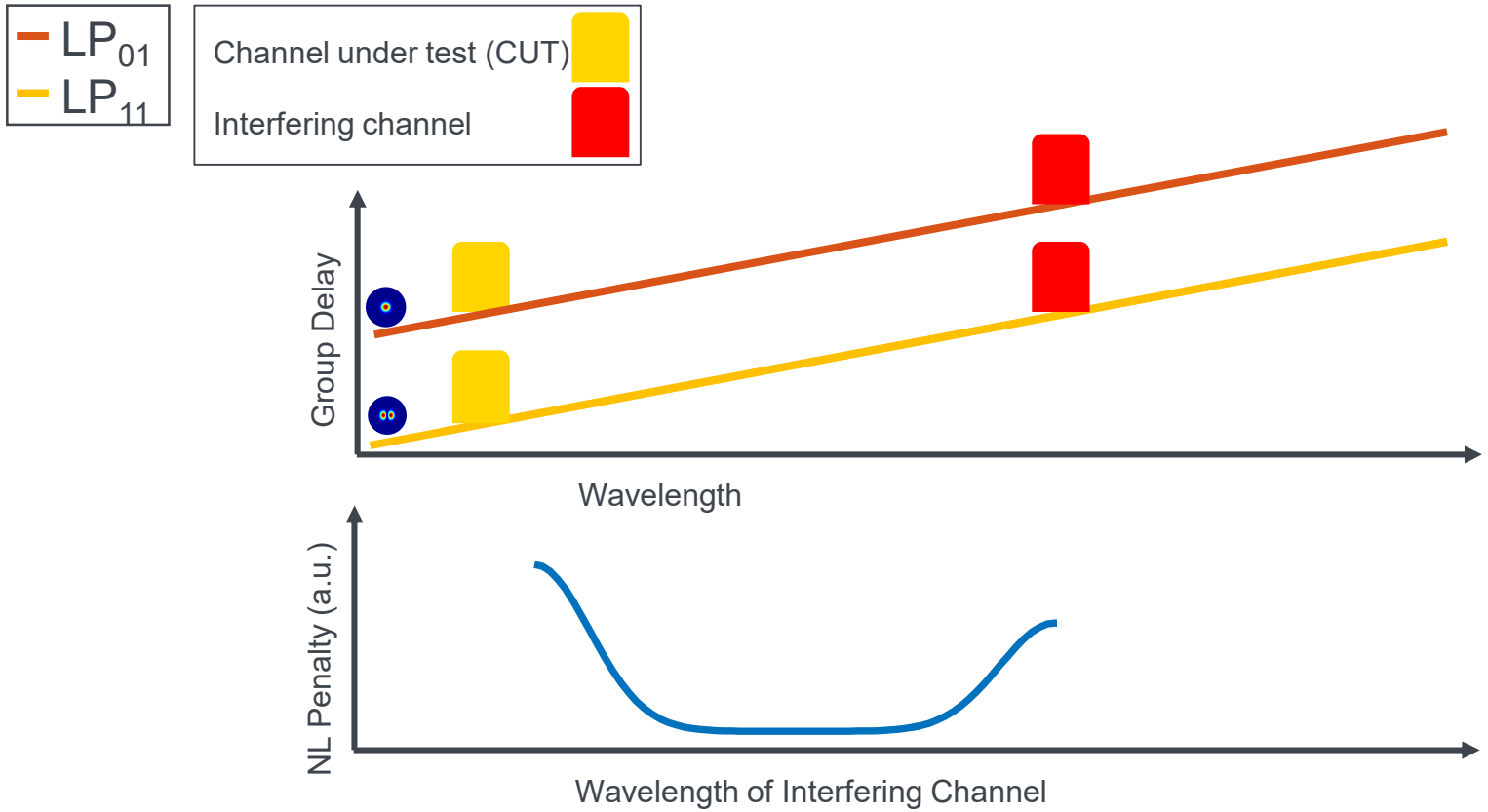
# Nichtlineare Signalverzerrung in einer standard Single-Mode Faser



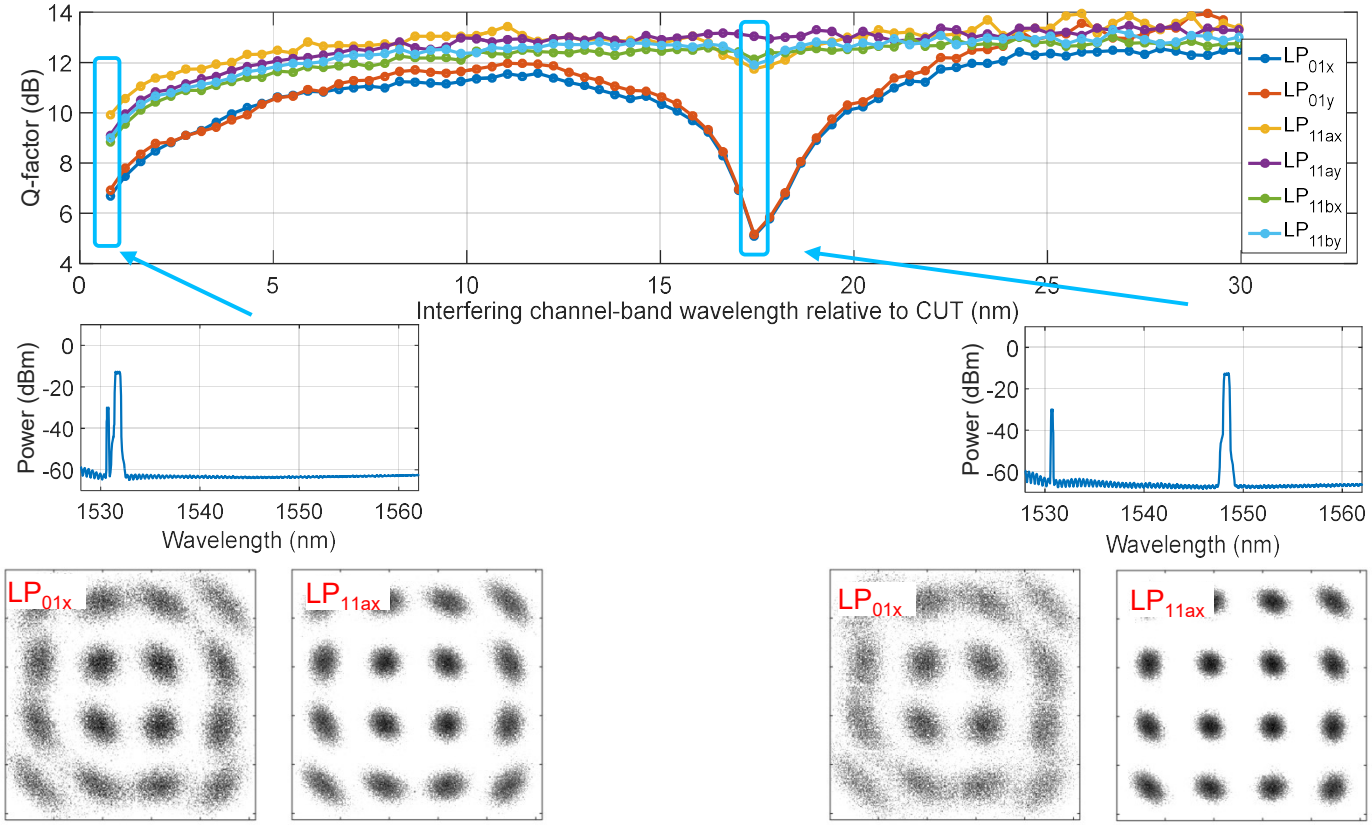
# Nichtlineare Signalverzerrung in einer Few-Mode Faser



# Nichtlineare Signalverzerrung in einer Few-Mode Faser



# Experimentale Demonstration




G. Rademacher et al., “Investigation of Intermodal Nonlinear Signal Distortions in Few-Mode Fiber Transmission”, Journal of Lightwave Technology vol. 37, 2019

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# Thank you!



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